

Aritro Roy

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Education

Asansol Engineering College, B.Tech in Computer Science
CGPA: 8.09/10

Mar 2022 – Jul 2026

Experience

SDE – Whatbytes Technologies

Oct 2025 – Present

- Architected a **FastAPI** multi-tenant document intelligence platform for CRE leases, with webhook-driven sync across Dropbox, Box, GDrive, and SharePoint (**Nango**, **GCS**) and a **Vertex AI** OCR → extraction pipeline structuring **100+ key points per lease** into isolated **Supabase** schemas.
- Reduced end-to-end ingestion latency by **60%** by parallelizing **Celery** async stages and optimising schema throughput.
- Designed **Airbyte** CDK ETL connectors (full + incremental syncs, **20+ SaaS entities**) and a **Django REST** backend for an AI red-teaming platform, featuring zero-code YAML tool registration and dependency-graph onboarding.
- Engineered a **Stripe**-backed billing engine (**Django REST**) using manual-capture PaymentIntents, atomic roll-back, and row-level locking (**select_for_update**), eliminating double-spend and duplicate-redemption races across all purchase flows.
- Built a template-driven notification system spanning **33 event types** and an admin-enforced variable allowlist, wiring cron-driven, time-offset triggers across bookings, wallet, referrals, and payouts.

Backend Engineering Intern – UrbanRider Technologies

Jul 2022 – May 2023

- Resolved **PostgreSQL** query bottlenecks and REST API inefficiencies, cutting application load latency by **22%**; restructured indexing to sustain performance under peak traffic.
- Eliminated manual release steps by automating validation, regression testing, and CI/CD with **Python**, **Docker**, and **GitHub Actions**, eradicating environment-parity bugs across staging and production.

Projects

Voice-Enabled RAG Pipeline with Measured Guardrails [\[Live\]](#) [\[GitHub\]](#) Python, FastAPI, FAISS, AWS

- Shipped a live voice-to-answer **RAG** service over a **99K-passage** corpus (**ElevenLabs** STT → **FAISS** → cross-encoder rerank → **Claude**) on **AWS EC2**, holding retrieval, reranking, and three guardrails to a **P50 of 96ms** against a 200ms budget — **26% faster** after removing a redundant re-embed (**111ms** → **12ms**).
- Benchmarked **8 chunking strategies** and **BM25/RRF** hybrid retrieval with paired bootstrap 95% CIs over **500 labelled queries**, proving ranking, not chunking, was the bottleneck; cross-encoder reranking lifted **recall@5 0.848** → **0.916**, and per-sentence groundedness guards caught **100% of ungrounded answers at zero false refusals**.

AgentTuring – Agentic API Routing System

[\[GitHub\]](#)

Python, FastAPI, Docker, Qdrant

- Engineered a **FastAPI** query router dispatching requests across **Qdrant** vector and structured data stores, with tool-use agentic workflows and async concurrency for multi-user workloads.
- Improved semantic query accuracy by **25%** over keyword baseline via dense vector retrieval, structured metadata filtering, and agentic re-ranking.

Technical Skills

Languages: Python, SQL, JavaScript

Frameworks & Tools: FastAPI, Django, DRF, Celery, PyTorch, Sentence Transformers, Airbyte CDK, Nango, Langfuse, Pytest

Data Engineering & AI: RAG Pipelines, ElevenLabs STT, Vertex AI, FAISS, Cross-Encoder Reranking, Hybrid Search (BM25 + RRF), Embeddings, ETL, Schema Design

Databases: PostgreSQL, Supabase, MongoDB, MySQL, Redis, Qdrant, Chroma DB, Pinecone DB

Cloud & DevOps: AWS, Docker, GitHub Actions, Caddy